

Week 9 — 🎉 The Caesar Cipher

Name: ____ Date: ____

Encode: slide each letter **forward** by the shift. Fall off at Z? Wrap to A — the alphabet is a **26-hour clock**, and $\text{mod}(\dots, 26)$ makes it tick.

Today's words

Word	What it means
cipher	a rule for scrambling a message
shift (the key)	how far each letter slides — the secret
wraparound / $\text{mod}(\dots, 26)$	past Z, reappear at A — remainder = clock math
brute force	crack a code by trying every possible key (all 25 of them)

Your cipher strip ✂️

A	B	C	D	E	F	G	H	I	J	K	L	M
0	1	2	3	4	5	6	7	8	9	10	11	12
N	O	P	Q	R	S	T	U	V	W	X	Y	Z
13	14	15	16	17	18	19	20	21	22	23	24	25

1 • Encode by hand 🔒 (shift 3, forward)

Slide each letter of **MATH** forward 3 places:

M → ____ A → ____ T → ____ H → ____

2 • Decode by hand 🔓 (shift 3, backward)

Someone sent you **G R J**. Slide each letter *back* 3 places:

G → ____ R → ____ J → ____ — the message is a: ____

3 • Clock math 🕒

Work these like the computer would:

- $\text{mod}(10 + 4, 12) = \underline{\hspace{2cm}}$
- $\text{mod}(25 + 3, 26) = \underline{\hspace{2cm}} \rightarrow$ **which letter is that?** ____
- $\text{mod}(23 + 5, 26) = \underline{\hspace{2cm}}$
- $\text{mod}(26, 26) = \underline{\hspace{2cm}}$
- Encoding used shift 8. The decode shift (forward!) is $26 - 8 = \underline{\hspace{2cm}}$

🧠 Brain teaser (optional — take it home)

Crack this ciphertext with your strip:

DTZ HWFHPJI YMJ HTIJ

Hint: the key equals the number of letters in the word SHIFT. Too easy? Then also try it the spy way: ignore the hint and brute-force it — but notice the three-letter word ending in **J**... a lot of English sentences have a three-letter word ending in E. Bring your solution (and your method!) next week for a shout-out.